

Activity: unit deviance

Unit deviance

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

The **unit deviance** is

$$d(y, \mu) = 2[t(y, y) - t(y, \mu)]$$

where $t(y, \mu) = y\theta - \kappa(\theta)$. This allows us to write the probability function in **dispersion model form**:

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

Poisson distribution

Suppose $Y \sim \text{Poisson}(\mu)$. We have previously verified that the Poisson is an EDM, with $\theta = \log(\mu)$, $\kappa(\theta) = \mu$, and $\phi = 1$.

1. Find the unit deviance $d(y, \mu)$ for the Poisson.

Negative binomial distribution

Suppose that $Y \sim NB(\mu, r)$, where r is a **known** constant. Then, the probability function for Y is

$$f(y; \mu, r) = \frac{\Gamma(y+r)}{\Gamma(y+1)\Gamma(r)} \left(\frac{\mu}{\mu+r}\right)^y \left(1 - \frac{\mu}{\mu+r}\right)^r$$

2. Show that the negative binomial (with known r) is an EDM, by identifying ϕ , θ , and $\kappa(\theta)$. **Note:** r is a known value – it does *not* have to be part of μ or ϕ !

3. Find the unit deviance $d(y, \mu)$ for the negative binomial.